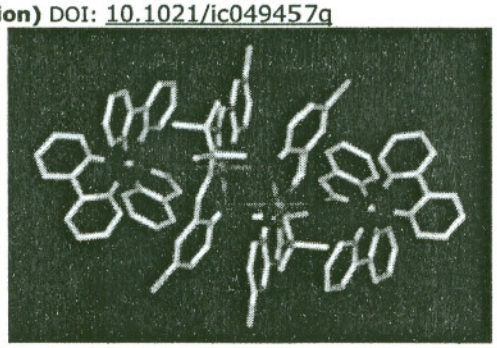
Inorganic Cumulative Exam
November 18, 2004

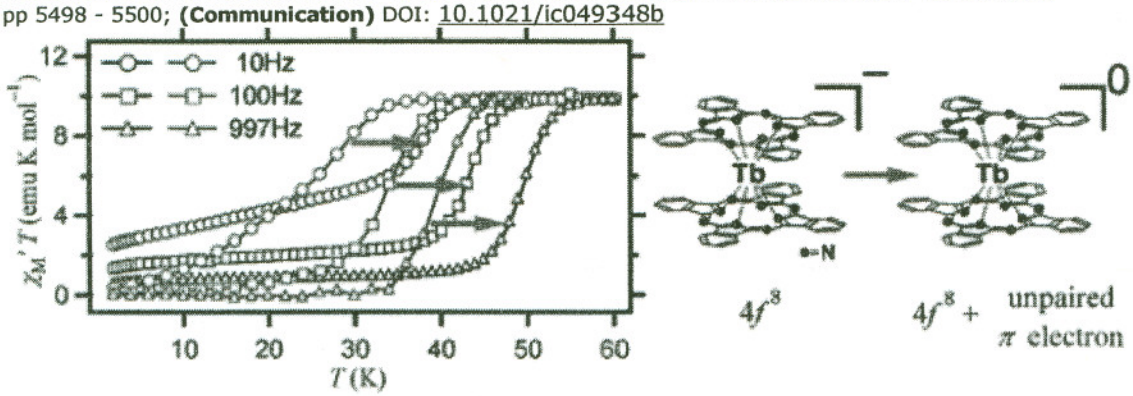
Today's chemical literature is massive and it is difficult to stay current with even a small area. However, inorganic chemistry students should develop a familiarity with the current "hot" topics and be able to spot papers that address these topics. Typically, authors will try to "sell" there research in terms of one of these topics in the introductory paragraphs. Below you will find the titles of 6 papers from the September issue of Inorganic Chemistry. Pick three and describe what the authors are doing and what important "hot" topic are they trying to address.

Linear Ni^{II}-Mn^{III}_2-Ni^{II} Tetramers: An Oligomeric Component of the $Mn^{III}_2Ni^{II}$ Single-Chain Magnets
Hitoshi Miyasaka, Tomohiro Nezu, Kunihisa Sugimoto, Ken-ichi Sugiura, Masahiro Yamashita, and Rodolphe Clérac
pp 5486 - 5488; **(Communication)** DOI: 10.1021/ic049457q



Abstract Full: HTML / PDF (69K) Supporting Info

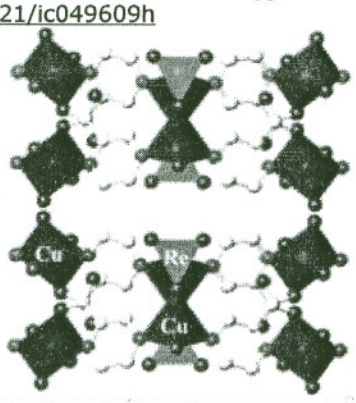
Upward Temperature Shift of the Intrinsic Phase Lag of the Magnetization of Bis(phthalocyaniato)terbium by Ligand Oxidation Creating an $S = {}^1/_2$ Spin
Naoto Ishikawa, Miki Sugita, Naohiro Tanaka, Tadahiko Ishikawa, Shin-ya Koshihara, and Youkoh Kaizu
pp 5498 - 5500; **(Communication)** DOI: 10.1021/ic049348b



Abstract Full: HTML / PDF (112K)

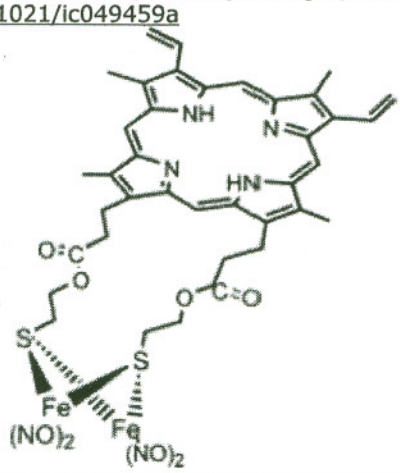
Synthesis and Characterization of ReO_4-Containing Microporous and Open Framework Structures

Junhua Luo, Ben Alexander, Timothy R. Wagner, and Paul A. Maggard
pp 5537 - 5542; **(Article)** DOI: 10.1021/ic049609h



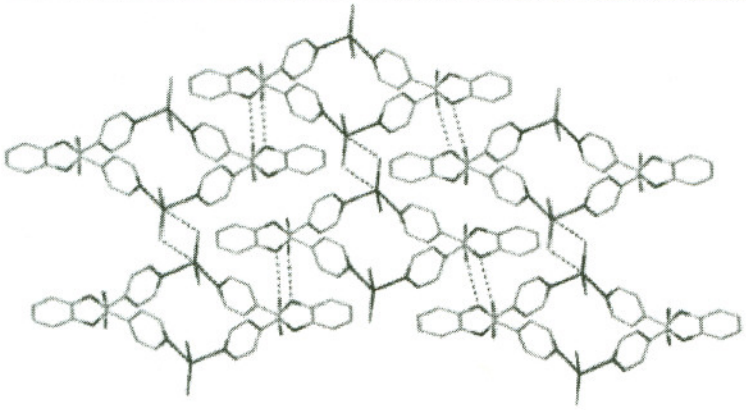
Abstract Full: HTML / PDF (396K) Supporting Info

Synthesis and Photochemical Properties of a Novel Iron-Sulfur-Nitrosyl Cluster Derivatized with the Pendant Chromophore Protoporphyrin IX[1]
Christa L. Conrado, Stephen Wecksler, Christian Egler, Douglas Magde, and Peter C. Ford
pp 5543 - 5549; **(Article)** DOI: 10.1021/ic049459a



Abstract Full: HTML / PDF (147K)

Self-Assembly Using Dynamic Coordination Chemistry and Hydrogen Bonding: Mercury(II) Macrocycles, Polymers and Sheets
Tara J. Burchell, Dana J. Eisler, and Richard J. Puddephatt
pp 5550 - 5557; **(Article)** DOI: 10.1021/ic049500+

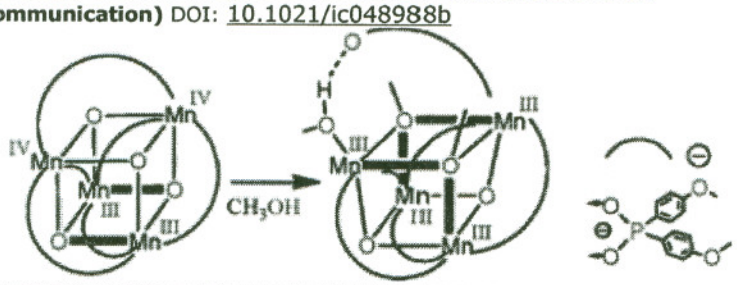


Abstract Full: HTML / PDF (361K) Supporting Info

Trapping an Elusive Intermediate in Manganese-Oxo Cubane Chemistry

Jian-Zhong Wu, Eric Sellitto, Glenn P. A. Yap, John Sheats, and G. Charles Dismukes

pp 5795 - 5797; **(Communication)** DOI: 10.1021/ic048988b



Abstract Full: HTML / PDF (75K) Supporting Info